1. Clinical Question & Core Hypothesis

Clinical Question

The path links inflammatory signaling (S100A9) to PD-L1–mediated immune suppression, raising the question why some tumors with abundant lymphocytes nevertheless fail to respond to PD-1/PD-L1–targeted therapy. 2) Tier 1 path used: the core S100A9 → PD-L1 axis (signaling network → S100A9 → PD-L1) because it provides a direct mechanistic route by which upstream inflammation could sustain checkpoint ligand expression and blunt T cell function. 3) Tier 2/supporting evidence: EPB41L2, SPP1 (Osteopontin) and FGFR3 link the risk-factor/inflammatory milieu to prognosis and may modulate myeloid or stromal contexts that reinforce S100A9-driven immunosuppression; these are retained as secondary modifiers. 4) Endpoint choice rationale: I selected therapy_response (PD-1/PD-L1 immune checkpoint blockade response) because PD-L1 is the proximal functional mediator in the path and the clinical puzzle concerns immune-evasion despite lymphocyte presence. 5) Deliberate exclusions: I excluded other imaging features (TLS-like aggregates, TSR) and placed EPB41L2/SPP1/FGFR3 in secondary_modifiers to keep the hypothesis sharply focused on the mechanistic question of how S100A9-driven PD-L1 expression functionally decouples TIL abundance from checkpoint therapy benefit?

Working Hypothesis

In hepatocellular carcinoma, high intratumoral S100A9 drives myeloid-cell–mediated paracrine upregulation and maintenance of CD274 (PD-L1) expression, thereby functionally uncoupling Tumor-Infiltrating Lymphocyte (TIL) density from effective cytotoxic T cell activity. Consequently, tumors with high TIL density and elevated S100A9/CD274 will show reduced therapy_response to PD-1/PD-L1 blockade compared with tumors that are TIL-high but S100A9-low or CD274-low. Secondary modifiers EPB41L2, SPP1, and FGFR3 are expected to modulate the magnitude of this effect by shaping the myeloid/stromal milieu that sustains PD-L1–mediated suppression. The hypothesis yields testable predictions: (a) in PD-1/PD-L1–treated HCC cohorts, the interaction of high TIL density with high S100A9 and CD274 expression predicts poorer therapy_response; (b) disrupting S100A9 signaling or PD-L1 reduces this resistance; (c) EPB41L2, SPP1, and FGFR3 status will stratify response within the TIL-high/S100A9/CD274-high category.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	S100A9, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

S100A9: S100A9 → PD-L1 axis (signaling network → S100A9 → PD-L1) because it provides a direct mechanistic route by which upstream inflammation could sustain checkpoint ligand expression and blunt T cell function.

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	EPB41L2	Supporting modifier from knowledge graph
Secondary	SPP1	Supporting modifier from knowledge graph
Secondary	FGFR3	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: expression often correlates with immune infiltration but does not universally predict benefit from PD-1/PD-L1 blockade, indicating additional regulatory layers. S100A9 is a damage-associated molecular pattern protein/alarmin that drives myeloid-cell recruitment and MDSC polarization, and can induce PD-L1 upregulation through TLR/NF-κB–type signaling in myeloid and possibly tumor cells.

Prediction 2: anchor the core mechanism; imaging feature (TIL density) captures the inflammatory context; and therapy_response as the endpoint aligns with clinical relevance in HCC treated with checkpoint inhibitors. Secondary modifiers EPB41L2, SPP1, and FGFR3 are plausible modulators of myeloid/stromal interactions and could refine the predictive model.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EPB41L2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SPP1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.